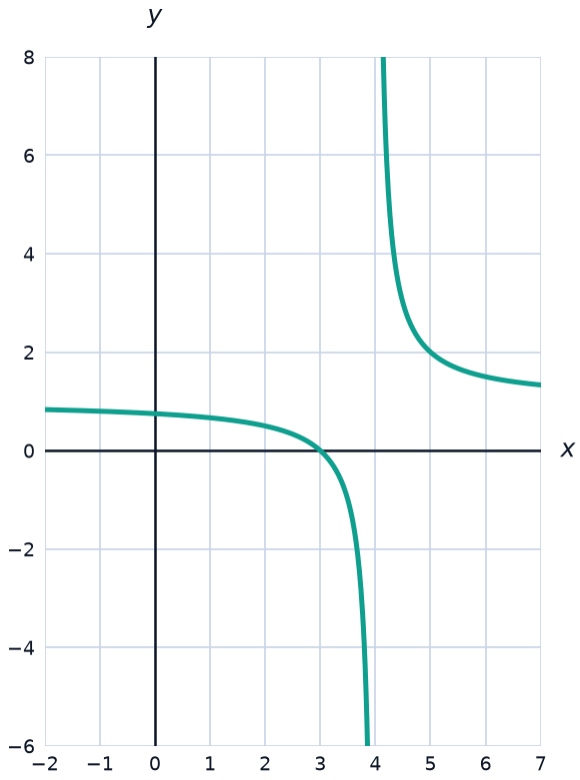
Rational Functions & Asymptotes

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vertical asymptotes and holes**

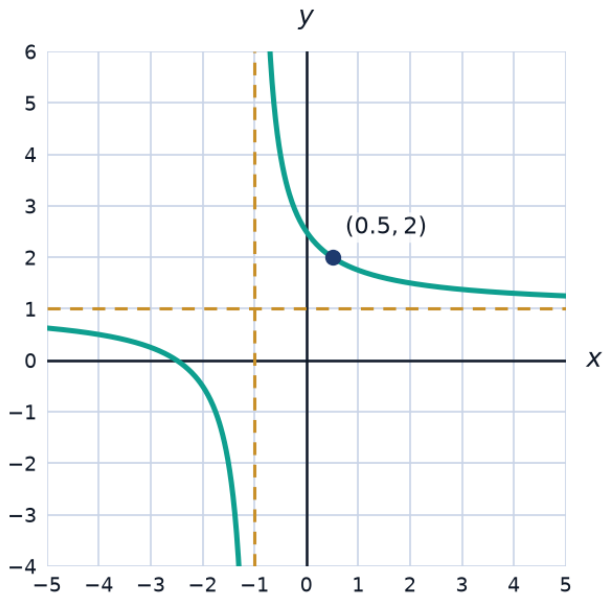
Find the vertical asymptotes and holes of $f(x) = \frac{x^2 - 9}{x^2 - x - 12}$.



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2 [Foundational] • Target: Vertical asymptotes and holes

Find the y-coordinate of the hole in the function from the previous problem.



3 [Foundational] • Target: Horizontal and slant asymptotes

Find the horizontal asymptote of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 4}$, using the degrees of numerator and denominator.

4 [Foundational] • Target: Horizontal and slant asymptotes

Find the horizontal asymptote of $g(x) = \frac{2x + 7}{x^2 - 1}$.

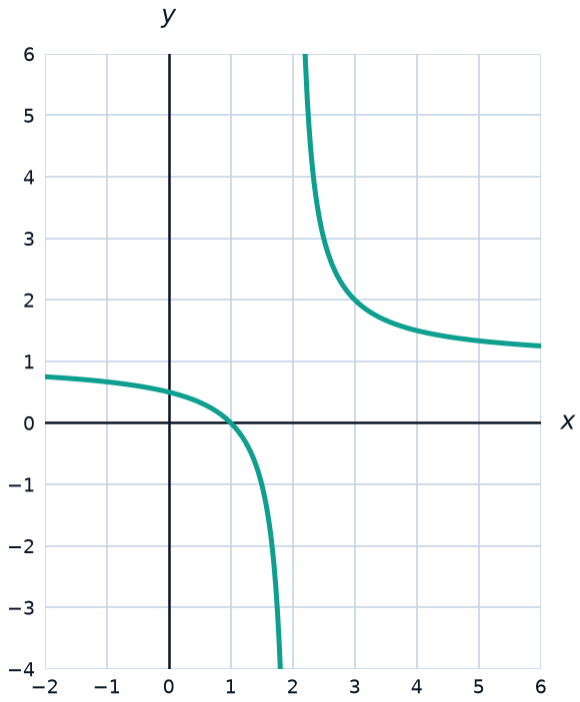
5 [Foundational] • Target: Horizontal and slant asymptotes

Explain why $h(x) = \frac{x^2 + 1}{x - 2}$ has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.

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6 [Foundational] • Target: Visualizing a rational function's asymptotes

The graph shows $f(x) = \frac{1}{x-2} + 1$, a transformed reciprocal function.



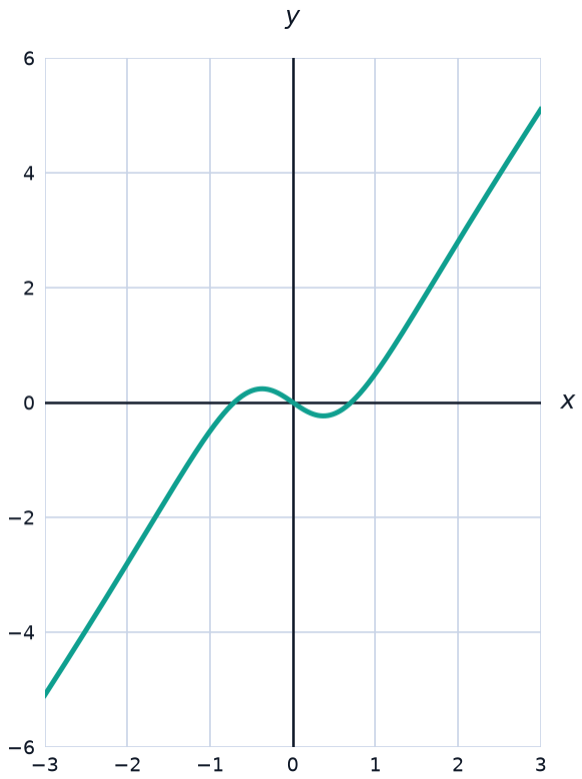
a Identify the vertical asymptote shown.

b Identify the horizontal asymptote shown.

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7 [Foundational] • Target: End behavior of rational functions

Describe the end behavior of $f(x) = \frac{2x^3 - x}{x^2 + 1}$ as $x \to \pm\infty$ (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).

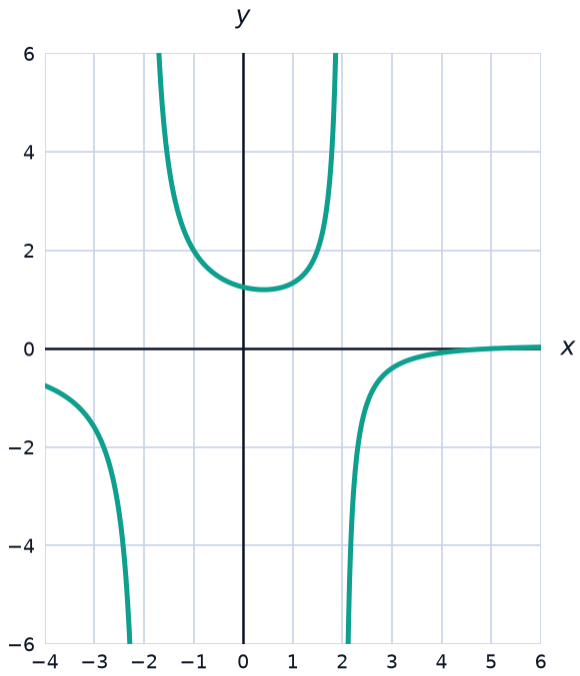


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Domain and intercepts

Find the domain, x-intercept(s), and y-intercept of $f(x) = \frac{x-5}{x^2-4}$.

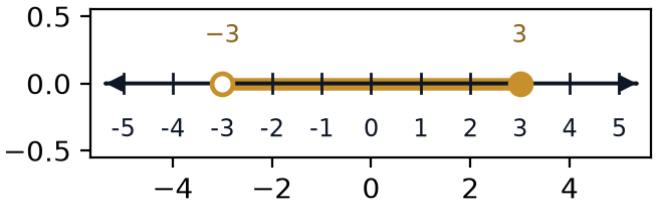


9 [Intermediate] • Target: Solving rational equations and inequalities

Solve $\frac{x+1}{x-3} = 2$, checking that the solution doesn't create a zero denominator.

10 [Intermediate] • Target: Solving rational equations and inequalities

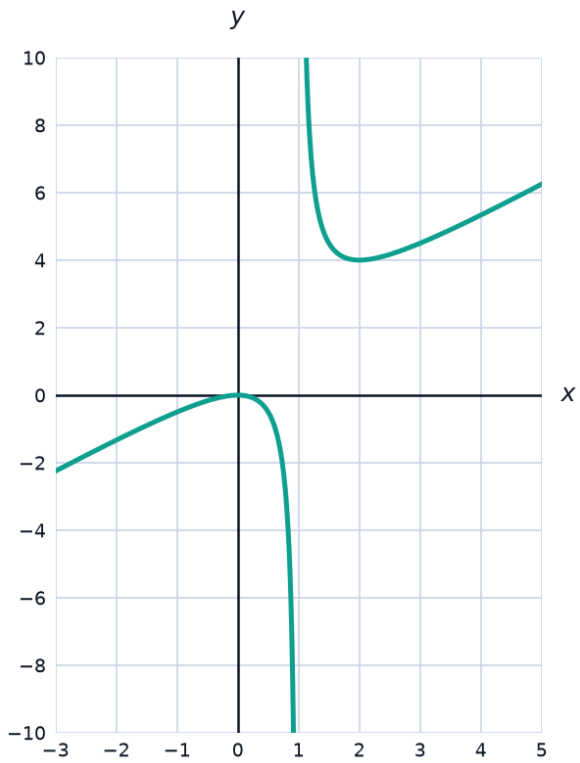
Solve the inequality $\frac{x-1}{x+2} > 0$ by determining the sign of the expression on each interval created by the zero and the excluded value.



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11 **[Intermediate]** • **Target: Visualizing a function with a slant asymptote**

The graph shows $f(x) = \frac{x^2}{x-1}$, which has a slant asymptote.



a Find the vertical asymptote and confirm it matches the graph.

b Use polynomial division to find the slant asymptote equation.

c Explain why this function has no horizontal asymptote.

12 **[Intermediate]** • **Target: Building a rational function from given features**

Construct a rational function with a vertical asymptote at $x = 3$, a hole at $x = -1$, and a horizontal asymptote at $y = 2$.

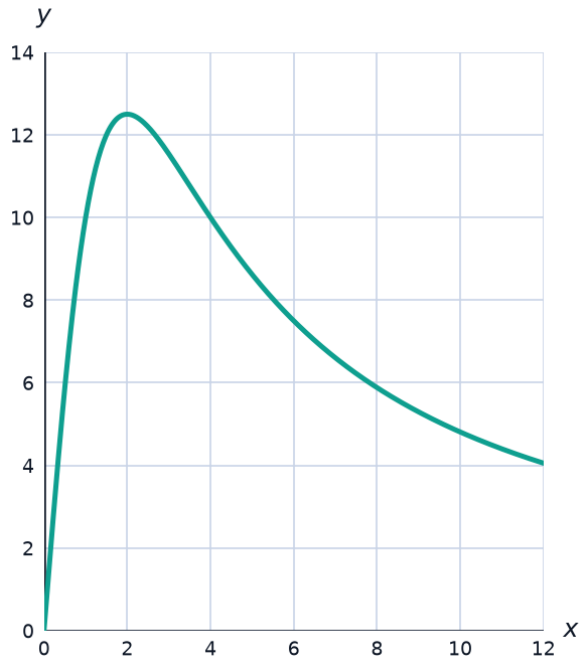
13 **[Intermediate]** • **Target: Building a rational function from given features**

Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

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14 [Intermediate] • Target: Rational functions in context

The concentration of a drug in the bloodstream t hours after a dose is modeled by $C(t) = \frac{50t}{t^2 + 4}$ (mg/L). Find the horizontal asymptote and interpret it in context.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

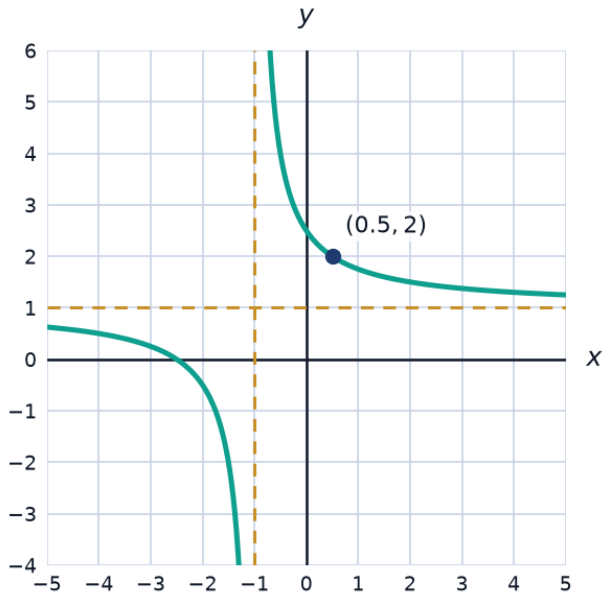
15 [Challenging] • Target: Horizontal and slant asymptotes • Cross-Lesson: 4.01 Inverse & Composite Functions

Let $f(x) = \frac{4e^{2x}}{e^{2x} + 3}$. Prove that f is strictly one-to-one and find an explicit formula for its inverse function $f^{-1}(x)$, stating the domain of f^{-1}.

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16 [Challenging] • Target: Visualizing a rational function's asymptotes • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Let $f(x) = 2x^2 - 3$ and $g(x) = \sqrt{x + 5}$. Find the composite function $(f \circ g)(x)$ and state its exact domain.



17 [Challenging] • Target: End behavior of rational functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

For a quadratic function $f(x)$, selected values are $f(1) = 4$, $f(2) = 9$, $f(3) = 18$, and $f(4) = 31$. Compute the first and second differences, verify that f is quadratic, and find its explicit formula $f(x) = ax^2 + bx + c$.

18 [Challenging] • Target: Domain and intercepts • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A bivariate dataset (x, y) is transformed into a semi-log plot $(x, \ln y)$, yielding the linear regression model $\ln(\hat{y}) = 1.5 + 0.4x$. Express $\hat{y}$ as an exponential function of x in the form $\hat{y} = a \cdot b^x$, and find the predicted percentage growth rate per unit increase in x.

19 [Challenging] • Target: Solving rational equations and inequalities • Cross-Lesson: 1.03 Rates of Change & Concavity

Prove the trigonometric identity $\frac{\sin(2x)}{1 + \cos(2x)} = \tan x$ and state the domain exclusions.

20 [Challenging] • Target: Visualizing a function with a slant asymptote • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Identify the conic section $4x^2 - 9y^2 - 16x - 18y - 29 = 0$ by completing the square. Find its center, vertices, foci, and asymptotes.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vertical asymptotes and holes

Find the horizontal asymptote of $f(x) = \frac{9x^2 + 5x - 2}{6x^2 + x - 7}$.

a. $y = \frac{2}{3}$

b. $y = \frac{5}{2}$

c. $y = 0$

d. $y = \frac{3}{2}$

22 [Foundational] • Target: Horizontal and slant asymptotes

Find the horizontal asymptote of $f(x) = \frac{2x^2 - 9x - 9}{2x^2 - 6x - 3}$.

a. $y = \frac{3}{2}$

b. $y = 0$

c. $y = 1$

d. $y = 2$

23 [Foundational] • Target: Visualizing a rational function's asymptotes

Find the horizontal asymptote of $f(x) = \frac{4x^2 - 8x + 6}{9x^2 + 3x + 9}$.

a. $y = \frac{9}{4}$

b. $y = \frac{4}{9}$

c. $y = 0$

d. $y = \frac{13}{9}$

24 [Foundational] • Target: End behavior of rational functions

Find the horizontal asymptote of $f(x) = \frac{6x^2 - 7x}{4x^2 - 7x - 9}$.

a. $y = \frac{3}{2}$

b. $y = \frac{2}{3}$

c. $y = 0$

d. $y = \frac{5}{2}$

25 [Foundational] • Target: Domain and intercepts

Find the horizontal asymptote of $g(x) = \frac{9x + 3}{7x^2 + 7x + 5}$.

a. $y = 0$

b. $y = 1$

c. $y = \frac{9}{7}$

d. $y = -\frac{9}{7}$

26 [Foundational] • Target: Solving rational equations and inequalities

Find the horizontal asymptote of $f(x) = \frac{9x^2 + 6x - 3}{8x^2 + 3x - 5}$.

a. $y = 0$

b. $y = \frac{9}{8}$

c. $y = \frac{8}{9}$

d. $y = \frac{17}{8}$

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27 [Foundational] • Target: Visualizing a function with a slant asymptote

Find the horizontal asymptote of $g(x) = \frac{-7x+1}{4x^2-4x+4}$.

a. $y = \frac{7}{4}$

b. $y = 0$

c. $y = -\frac{7}{4}$

d. $y = 1$

28 [Foundational] • Target: Building a rational function from given features

Find the hole and vertical asymptote of $f(x) = \frac{(x+2)(x-5)}{(x+2)(x-6)}$.

a. Hole at $x = 6$; vertical asymptote at $x = -2$.

b. Hole at $x = -2$; vertical asymptote at $x = 6$.

c. Vertical asymptotes at $x = -2$ and $x = 6$, no hole.

d. Hole at $x = 5$; vertical asymptote at $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rational functions in context

Find the hole and vertical asymptote of $f(x) = \frac{(x-2)(x-4)}{(x-2)(x+6)}$.

a. Hole at $x = 2$; vertical asymptote at $x = -6$.

b. Vertical asymptotes at $x = 2$ and $x = -6$, no hole.

c. Hole at $x = -6$; vertical asymptote at $x = 2$.

d. Hole at $x = 4$; vertical asymptote at $x = -6$.

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30 [Intermediate] • Target: Vertical Asymptotes vs. Removable Discontinuities

Find the hole and vertical asymptote of $f(x) = \frac{(x+8)(x-8)}{(x+8)(x-3)}$.

a. Vertical asymptotes at $x = -8$ and $x = 3$, no hole.

b. Hole at $x = -8$; vertical asymptote at $x = 3$.

c. Hole at $x = 8$; vertical asymptote at $x = 3$.

d. Hole at $x = 3$; vertical asymptote at $x = -8$.

31 [Intermediate] • Target: Horizontal Asymptotes and Degrees

Find the hole and vertical asymptote of $f(x) = \frac{(x+8)(x-8)}{(x+8)(x-9)}$.

a. Hole at $x = -8$; vertical asymptote at $x = 9$.

b. Vertical asymptotes at $x = -8$ and $x = 9$, no hole.

c. Hole at $x = 8$; vertical asymptote at $x = 9$.

d. Hole at $x = 9$; vertical asymptote at $x = -8$.

32 [Intermediate] • Target: Slant (Oblique) Asymptotes

Find the hole and vertical asymptote of $f(x) = \frac{(x-1)(x+1)}{(x-1)(x)}$.

a. Hole at $x = 1$; vertical asymptote at $x = 0$.

b. Vertical asymptotes at $x = 1$ and $x = 0$, no hole.

c. Hole at $x = 0$; vertical asymptote at $x = 1$.

d. Hole at $x = -1$; vertical asymptote at $x = 0$.

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33 **[Intermediate]** • **Target: Limit Notation and Asymptotic Behavior**

Find the hole and vertical asymptote of $f(x) = \frac{(x-6)(x+1)}{(x-6)(x-5)}$.

a. Hole at $x = -1$; vertical asymptote at $x = 5$.

b. Vertical asymptotes at $x = 6$ and $x = 5$, no hole.

c. Hole at $x = 6$; vertical asymptote at $x = 5$.

d. Hole at $x = 5$; vertical asymptote at $x = 6$.

34 **[Intermediate]** • **Target: Vertical asymptotes and holes**

Find the hole and vertical asymptote of $f(x) = \frac{(x+6)(x+4)}{(x+6)(x+7)}$.

a. Vertical asymptotes at $x = -6$ and $x = -7$, no hole.

b. Hole at $x = -4$; vertical asymptote at $x = -7$.

c. Hole at $x = -7$; vertical asymptote at $x = -6$.

d. Hole at $x = -6$; vertical asymptote at $x = -7$.

35 **[Intermediate]** • **Target: Horizontal and slant asymptotes**

Find the slant asymptote of $h(x) = \frac{x^2+8x-7}{x+9}$.

a. $y = x$

b. $y = x - 2$

c. $y = x + 1$

d. $y = x - 1$

36 **[Intermediate]** • **Target: Visualizing a rational function's asymptotes**

Find the slant asymptote of $h(x) = \frac{x^2+4x+5}{x+6}$.

a. $y = x - 3$

b. $y = x - 1$

c. $y = x - 2$

d. $y = x + 2$

37 **[Intermediate]** • **Target: End behavior of rational functions**

Solve $\frac{x+4}{x-3} = -3$ for x.

a. $-\frac{5}{4}$

b. $\frac{5}{4}$

c. $-\frac{4}{3}$

d. $\frac{9}{4}$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Domain and intercepts • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve $\frac{x+3}{x} = -2$ for x.

a. 1

b. $-\frac{3}{2}$

c. 0

d. -1

39 [Challenging] • Target: Solving rational equations and inequalities • Cross-Lesson: 1.03 Rates of Change & Concavity

Solve the inequality $\frac{x-6}{x-7} > 0$.

a. $(-\infty, 6) \cup (7, \infty)$

b. $[6, 7]$

c. $(6, 7)$

d. $(-\infty, 7) \cup (6, \infty)$

40 [Challenging] • Target: Visualizing a function with a slant asymptote • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Solve the inequality $\frac{x-2}{x-7} > 0$.

a. $(-\infty, 7) \cup (2, \infty)$

b. $(2, 7)$

c. $(-\infty, 2) \cup (7, \infty)$

d. $[2, 7]$

41 [Challenging] • Target: Vertical Asymptotes vs. Removable Discontinuities • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find all vertical asymptotes and holes for $f(x) = \frac{x^2-9}{x^2-x-6}$.

a. Vertical asymptote at $x = -2$; hole at $x = 3$

b. Vertical asymptote at $x = 3$; hole at $x = -2$

c. Vertical asymptotes at $x = 3$ and $x = -2$; no holes

d. Holes at $x = 3$ and $x = -2$; no vertical asymptotes

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42 [Challenging] • Target: Vertical Asymptotes vs. Removable Discontinuities • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

If $h(x) = \frac{x-4}{(x-4)^2(x+1)}$, which describes the behavior of h near $x = 4$?

a. $h(x)$ has a vertical asymptote at $x = 4$

b. $h(x)$ has a removable hole at $x = 4$

c. $h(x)$ has a horizontal tangent at $x = 4$

d. $h(x)$ is continuous at $x = 4$

43 [Challenging] • Target: Horizontal Asymptotes and Degrees • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Can the graph of a rational function intersect its horizontal asymptote?

a. Yes, the graph can cross a horizontal asymptote at finite values of x

b. No, an asymptote can never be touched or crossed by definition

c. Only if the horizontal asymptote is $y = 0$

d. Only if the function has no vertical asymptotes

44 [Challenging] • Target: Slant (Oblique) Asymptotes • Cross-Lesson: 4.01 Inverse & Composite Functions

Under what condition does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

a. The degree of $P(x)$ is exactly 1 greater than the degree of $Q(x)$

b. The degree of $P(x)$ equals the degree of $Q(x)$

c. The degree of $Q(x)$ is exactly 1 greater than the degree of $P(x)$

d. The degree of $P(x)$ is at least 2 greater than the degree of $Q(x)$

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45 **[Challenging]** • **Target: Limit Notation and Asymptotic Behavior** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

For $f(x) = \frac{1}{x-3}$, what are $\lim_{x \to 3^+} f(x)$ and $\lim_{x \to 3^-} f(x)$?

a. $\lim_{x \to 3^+} f(x) = \infty$ and $\lim_{x \to 3^-} f(x) = -\infty$

b. $\lim_{x \to 3^+} f(x) = -\infty$ and $\lim_{x \to 3^-} f(x) = \infty$

c. Both limits equal ∞

d. Both limits equal $-\infty$

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